

$y = b^x$ is a one-to-one function that has an inverse.

Using: $f^{-1}(x) = y \Leftrightarrow x = f(y)$, we have that

$$\log_b x = y \Leftrightarrow b^y = x$$

Cancellation equations:

$$\log_b(b^x) = x \text{ for every } x \in \mathbb{R} \quad (\text{p. 58})$$

$$b^{\log_b x} = x \text{ for every } x > 0.$$

Graphs: see p. 58

Laws of Logarithms: .58

Do example 6

Natural Logarithm: $\log_e x = \ln x$.

$$\ln x = y \Leftrightarrow e^y = x$$

MR Examples 7, 8, 9 on p. 59-60.

Change of base formula: For any $b > 0$ ($b \neq 1$), we have

$$\log_b x = \frac{\ln x}{\ln b}.$$

Laws: For $a, b > 0$ and $r \in \mathbb{R}$, we have:

$$1) \ln ab = \ln a + \ln b$$

$$2) \ln \frac{a}{b} = \ln a - \ln b$$

$$3) \ln a^r = r \ln a$$

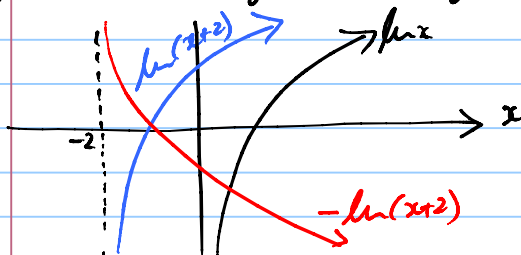
$$\log_e e = \ln e = 1.$$

$$4) \ln e^x = x \quad \forall x \in \mathbb{R}$$

$$5) e^{\ln x} = x \quad \forall x > 0$$

Examples:

1) Sketch the graphs of $y = \ln(x+2)$ and $y = -\ln(x+2)$



$$2) \ln e = \ln 1 - \ln e = 0 - 1 = -1.$$

$$\text{or } \ln \frac{1}{e} = \ln e^{-1} = -1 \ln e = -1.$$

$$3) e^{-2\ln 5} = e^{\ln 5^{-2}} = 5^{-2} = \frac{1}{25}.$$

$$4) \ln \left(\frac{x^3 e^{x^2}}{\sqrt{x}} \right)$$

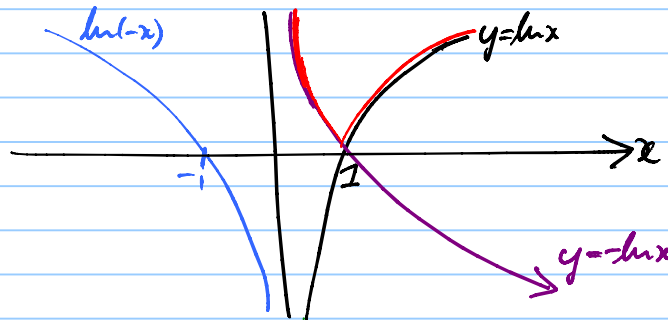
$$= \ln x^3 + \ln e^{x^2} - \ln \sqrt{x}$$

$$= 3 \ln x + x^2 \ln e - \ln x^{\frac{1}{2}}$$

$$= 3 \ln x + x^2 - \frac{1}{2} \ln x$$

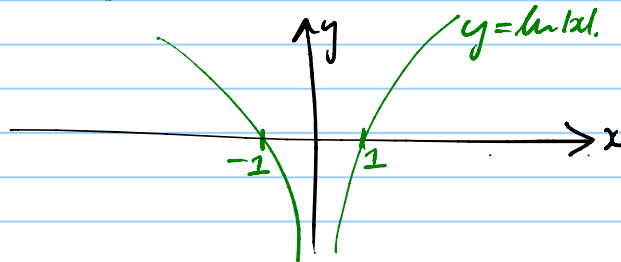
$$= \underline{\underline{\frac{5}{2} \ln x + x^2}}$$

5) Sketch: $y = \ln x$, $y = \ln(-x)$, $y = -\ln(x)$
 $y = |\ln x|$ and $y = \ln|x|$.



$$y = |\ln x| = \begin{cases} \ln x & \text{if } \ln x \geq 0 \\ -\ln x & \text{if } \ln x < 0 \end{cases}$$

$$y = \ln|x| = \begin{cases} \ln x & \text{if } x > 0 \\ \ln(-x) & \text{if } x < 0 \end{cases}$$



Examples: solving equations

$$1) 2 \ln x - \ln(x-1) = \ln 10 - \ln 2$$

$$\Rightarrow \ln x^2 - \ln(x-1) = \ln \frac{10}{2}$$

$$\Rightarrow \ln \frac{x^2}{x-1} = \ln 5$$

$$\Rightarrow \frac{x^2}{x-1} = 5$$

$$\Rightarrow x^2 = 5x - 5$$

$$\Rightarrow x^2 - 5x + 5 = 0 \Rightarrow x = \frac{5 \pm \sqrt{25 - 20}}{2}$$

$$= \frac{5 \pm \sqrt{5}}{2}$$

$$2) e^{5-3x} = 10$$

$$\Rightarrow \ln(e^{5-3x}) = \ln 10$$

$$\Rightarrow 5-3x = \ln 10$$

$$\Rightarrow 3x = 5 - \ln 10$$

$$x = \frac{1}{3}(5 - \ln 10)$$

$$3) e^2 = 9e^{3x}$$

$$\Rightarrow \frac{1}{9} = \frac{e^{3x}}{e^2}$$

$$\Rightarrow \frac{1}{9} = e^{3x-2} = e^{2x}$$

$$\Rightarrow \ln 9^{-1} = \ln e^{2x}$$

$$\Rightarrow -\ln 9 = 2x$$

$$\Rightarrow -\ln 3^2 = -2\ln 3 = 2x$$

$$\Rightarrow \underline{x = -\ln 3}$$

Solving inequalities

$$1) \ln x < 3$$

$$\Rightarrow e^{\ln x} < e^3 \quad (\text{why?? Remember } e^x \text{ is an increasing function})$$

$$\Rightarrow x < 3 \text{ and } x > 0, \text{ therefore } \underline{0 < x < 3}$$

$$2) 1 - \ln(x+1) > 2$$

$$\Rightarrow 1-2 > \ln(x+1), \quad x+1 > 0$$

$$\Rightarrow \ln(x+1) < -1 \text{ and } x > -1$$

$$\Rightarrow e^{\ln(x+1)} < e^{-1}$$

$$\Rightarrow x+1 < 1/e \text{ and } x > -1$$

$$\Rightarrow x < 1/e - 1 \text{ and } x > -1 \Rightarrow \underline{x \in (-1, -1 + \frac{1}{e})}$$

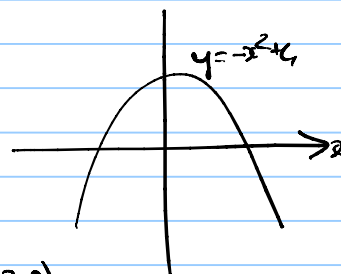
Domains

Find the domain of:

$$1) \ln(4-x^2)$$

$$\text{We thus have: } 4-x^2 > 0$$

and we use a graph to solve this:



$$\Rightarrow \text{From the graph, we have: } x \in (-2, 2)$$

$$(2) y = \frac{1}{\ln x}$$

$$\ln x \neq 0 \Rightarrow x \neq 1, \text{ since } \ln 1 = 0$$

$$\Rightarrow \text{Domain: } x \in (0, 1) \cup (1, \infty).$$

$$(3) y = \frac{1}{1 - \ln(x-1)}$$

$$1 - \ln(x-1) \neq 0$$

$$\Rightarrow \ln(x-1) \neq 1$$

$$\text{Suppose } \ln(x-1) = 1$$

$$\Rightarrow x-1 = e$$

$$\Rightarrow x = 1+e$$



$$\Rightarrow x-1 > 0 \text{ and } x \neq 1+e$$

$$\Rightarrow x > 1 \text{ and } x \neq 1+e$$

$$\Rightarrow x \in (1, 1+e) \cup (1+e, \infty).$$